

Towards Pre-emptive Physiological Error Classification Within Airborne Environments

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Abstract

Human error contributes to 70-80% of aviation accidents despite technological advances, yet current safety systems remain reactive. We present the first empirical validation that neurophysiological signals can predict pilot errors up to 10 seconds in advance during operational flight. Through trials with nine commercial pilots, we collected synchronised EEG and eye-tracking data across laboratory, straight-and-level flight, and 2G manoeuvre conditions whilst participants performed multitasking scenarios. EEG-based classifiers achieved peak F1 scores of 88.7% in controlled environments and 82.0% during flight operations, with eye-tracking providing complementary predictive patterns. Results demonstrate that physiological error precursors remain detectable despite motion artefacts, electromagnetic interference, and gravitational stress. These findings establish the feasibility of pre-emptive error monitoring in operational aviation environments, providing a foundation for cognitive-aware cockpit systems that could enable proactive safety interventions before errors occur.

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1 Introduction

Human error persists as the predominant factor in aviation accidents, accounting for 70-80% of incidents despite substantial technological and operational advancements [20, 40]. Current safety frameworks rely on post-incident analysis and self-reporting mechanisms that remain inherently reactive, providing valuable insights for systemic improvements but fundamentally unable to prevent errors in real-time [21, 30]. This reactive paradigm has motivated exploration of proactive approaches that could detect and potentially mitigate errors before they occur.

Laboratory research has demonstrated that the human brain provides measurable warning signs of impending errors. Foundational work by Gehring et al. [9] and Falkenstein et al. [8] revealed specific neural signatures detectable through electroencephalography (EEG) several seconds before behavioural error manifestation.

More recently, Estiveira et al. [7] showed that midfrontal theta activity increases significantly 2-4 seconds before error commission, suggesting a generalised neural mechanism for error anticipation independent of task domain. Eye-tracking research has identified complementary predictive signals, with characteristic changes in gaze patterns and visual scanning behaviours preceding errors by 1-3 seconds [3, 14]. Dehais et al. [4] demonstrated that gaze behaviours reliably signal imminent pilot errors, underscoring the relevance of oculomotor tracking for aviation. These physiological indicators offer a compelling possibility: detecting errors before they occur through continuous cognitive monitoring [5].

However, translating these laboratory findings to operational aviation environments presents substantial challenges. Dynamic flight conditions introduce motion artefacts, gravitational stressors, and electromagnetic interference that degrade physiological signal quality [16, 41]. Early field implementations revealed significant degradation in signal reliability compared to laboratory baselines [5], raising questions about whether error-specific markers remain detectable outside controlled settings. Moreover, existing research has largely focused on workload assessment rather than specific error prediction [11, 17, 18, 24?], and whilst post-error detection systems have achieved 75-80% accuracy in laboratory tasks [2, 42], prior work has only demonstrated initial feasibility of error classification from physiological signals in airborne environments [27]. No prior study has conducted a full empirical validation with multimodal pre-emptive detection across progressive flight conditions. This gap is critical: without evidence that predictive physiological markers remain detectable and actionable under real-world aviation conditions, the development of proactive safety systems remains speculative.

To address this gap, we conducted live flight trials with nine commercial pilots, employing a multimodal sensing framework integrating EEG, ECG, and eye-tracking during controlled cognitive tasks under varying environmental stressors. Participants performed standardised multitasking scenarios across three progressively complex conditions: controlled laboratory settings, straight-and-level flight, and demanding 2G spiral manoeuvres. We focused analysis on the 10-second window preceding error events, implementing a temporal buffer strategy to ensure classifiers learned genuine pre-error signatures rather than confounded post-error artefacts [39]. Two temporal analysis strategies were compared: discrete windowing (10 non-overlapping 1-second segments) to localise when predictive features emerge, and cumulative windowing (aggregated segments extending backwards from error onset) to simulate real-time system operation.



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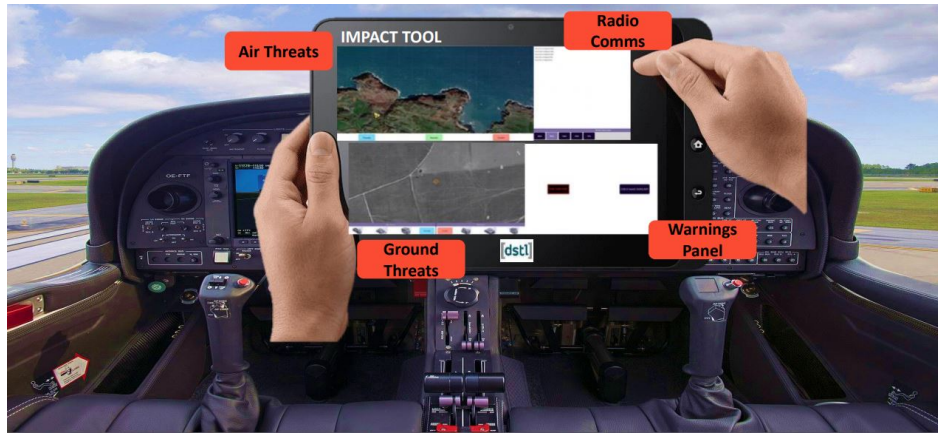


Figure 1: Visualisation of the IMPACT Tool interface used during flight trials. Participants completed four concurrent cognitive tasks while physiological signals were recorded. Image adapted from [36].

Our findings demonstrate that EEG signals retain high predictive validity, achieving peak F1 scores of 88.7% in controlled conditions and 82.0% in airborne environments, with error precursors detectable throughout the 10-second pre-error window—a substantial extension beyond the 2-4 second anticipation windows reported in previous laboratory studies [7]. Eye-tracking data provided complementary discriminative power (peak F1: 78.0%), particularly in the final seconds preceding errors, corroborating its role in monitoring attentional lapses [14]. Crucially, these physiological signatures remained robust despite environmental perturbations, including vibration, electromagnetic interference, and gravitational stress, establishing their feasibility for operational deployment.

This work provides three key advances: (1) the first empirical validation of pre-emptive error detection during live flight conditions, demonstrating that laboratory-derived approaches translate effectively to operational environments; (2) quantification of multimodal physiological indicators with extended temporal lead times, showing that error precursors emerge substantially earlier than previously documented; and (3) evidence that cumulative temporal integration improves prediction performance (average 3.2% gain for EEG), providing practical guidance for system implementation. By bridging the gap between laboratory research and operational aviation needs, our findings lay the foundation for cognitive-aware cockpit systems that could enable proactive safety interventions, addressing errors at their neurophysiological origin before behavioural manifestation [4, 32].

2 Methodology

2.1 Study Overview

Power analysis indicated a minimum requirement of eight participants for adequate statistical power (medium effect size $f=0.35$, power=0.80, $\alpha=0.05$). We recruited nine male commercial pilots holding UK Commercial Pilot Licenses with DA42 aircraft clearance. Due to prescription eyewear requirements during flight, eye-tracking data was collected from four participants. The study protocol was approved by the anonymised institute's ethics review board.

The experimental design implemented three progressive testing environments to evaluate physiological error detection under increasing operational complexity: *Baseline* (controlled laboratory free from flight-specific stressors), *Straight-and-Level flight (S&L)* (stable flight operations introducing vibration, electromagnetic interference, and confined spaces), and *2G* (sustained three-minute spiral manoeuvres generating twice normal gravitational force, introducing significant physical and vestibular demands). All flight scenarios were conducted in a Diamond DA-42NG aircraft between 3,000 and 8,000 feet in non-pressurised conditions [5].

2.2 Task Design and Data Collection

To elicit errors whilst capturing physiological responses, we employed the IMPACT Tool [36], a tablet-based multitasking simulation platform (Figure 1). The tool consists of four concurrent cognitive tasks: *Air Threats* (audio-visual target classification), *Radio Communications* (spatial tracking through auditory updates), *Ground Threats* (visual discrimination with time constraints), and *Warnings Panel* (system monitoring with 5-second response windows). Task difficulty was systematically varied across three levels (low, medium, high) by adjusting event frequency and time constraints, maintaining each difficulty for three-minute blocks. The IMPACT Tool automatically timestamps all participant interactions, providing precise labels for aligning physiological data with error events. To avoid class imbalance, we uniformly sampled error-free segments as baseline comparison samples across all conditions [11].

2.3 Experimental Procedure and Data Collection

Participants were fitted with research-grade physiological sensors: g.tec EEG system (24 channels, 256 Hz sampling rate), Tobii Pro Glasses 3 for eye-tracking (100 Hz sampling rate, $n=4$), and Polar H10 ECG chest strap (130 Hz sampling rate). All devices were CE-certified and cleared for use in flight environments. Following familiarisation with the IMPACT Tool, participants progressed through laboratory, straight-and-level, and 2G spiral conditions, with task difficulty levels randomised within each environment to

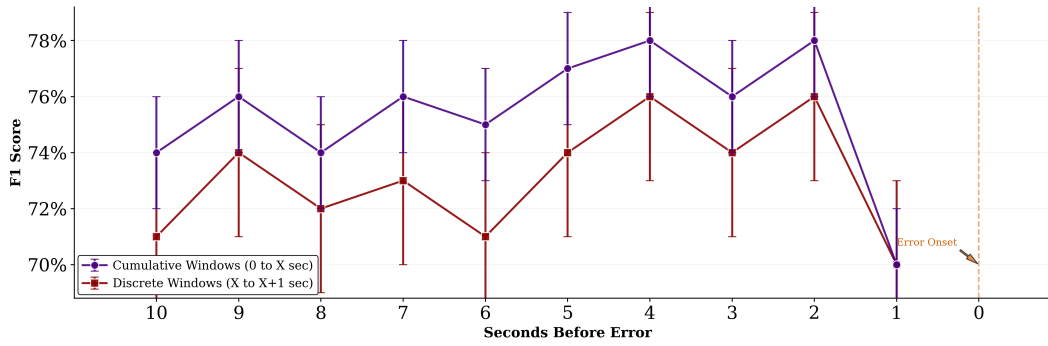


Figure 2: Eye-tracking-based classification F1 scores across the 10-second pre-error window, showing characteristic temporal variability with distinct performance peaks at 9, 7, 4, and 2 seconds pre-error.

mitigate order effects. All flight operations were conducted under strict safety protocols.

2.4 Data Processing and Classification

Aviation environments introduce substantial artefacts into physiological recordings, necessitating careful preprocessing [5]. EEG data underwent bandpass filtering (0.5-50 Hz) to remove muscle noise and electromagnetic artefacts [41], followed by average re-referencing to improve mobile signal quality [16]. Eye-tracking data were processed using I-VT fixation filtering for detecting saccades and fixations [14]. ECG signals were filtered using established protocols to isolate cardiac components [13, 23].

Feature extraction employed 1-second sliding windows with no overlap, capturing neurophysiological changes with sufficient temporal resolution for predictive modeling [2]. From EEG, we extracted features across six categories: frequency domain (power spectral density bands: delta, theta, alpha, beta, gamma), statistical (mean, variance, skewness, kurtosis), morphological (curve length, peak count, non-linear energy), time-frequency (wavelet transforms), linear (autoregressive coefficients), and non-linear measures (approximate entropy, Hurst exponent) [10, 12, 16, 19, 38, 41]. Eye-tracking features captured fixation metrics, saccadic amplitudes, pupillometry, and blink patterns [1, 6, 22]. ECG features focused on heart rate variability measures [37].

To test whether physiological signals could anticipate errors, we focused on the 10-second window preceding each error event [7]. We implemented a temporal buffer strategy that excluded error samples occurring within 10 seconds of prior errors, ensuring classifiers learned genuine pre-error signatures rather than confounded post-error artefacts [39]. Two temporal strategies were compared: **Discrete Windowing** (10 non-overlapping 1-second segments to localise when predictive features emerge) and **Cumulative Windowing** (aggregated segments extending backwards from error onset, simulating real-time system operation). For classification, we employed three established machine learning models: Support Vector Machine (RBF kernel), Random Forest [31], and Multi-layer Perceptron [34]. These models were selected for their demonstrated effectiveness in EEG-based cognitive state classification, including mental workload estimation [18, 24?]. Models were trained separately on individual physiological modalities (EEG, eye-tracking,

ECG) using balanced data. Performance was evaluated using F1-score with statistical significance determined against a 50% random baseline using t-tests ($p < 0.001$).

3 Results

3.1 Temporal Evolution of Error Prediction

EEG signals provided strong predictive capability throughout the 10-second pre-error window (Figure 3). The cumulative windowing approach achieved a peak F1 score of 88.7% at 9 seconds before error commission, with performance remaining significantly above chance ($p < 0.001$) across all timepoints. Only a minor decline occurred immediately prior to error execution, supporting the notion that stable neural indicators emerge well before behavioural lapses [7].

Eye-tracking measures demonstrated more temporal variability (Figure 2), achieving the highest F1 score of 78.0% at 2 seconds pre-error, with additional characteristic peaks at 9, 7, and 4 seconds before error onset. These temporal markers correspond to theorised attentional shifts during error formation [14]. Although performance was lower than EEG overall, eye-tracking predictions remained statistically significant ($p < 0.05$) across most timepoints, reinforcing their complementary role.

3.2 Comparative Analysis of Temporal Strategies

Cumulative windowing consistently outperformed discrete windowing across both modalities. For EEG, cumulative integration yielded an average 3.2% improvement in F1 score, with largest gains observed between 5-7 seconds pre-error. This suggests that aggregating physiological information over time better captures evolving cognitive processes leading to error than isolated momentary snapshots. Discrete windowing, despite greater variability, revealed important insights into temporal emergence of predictive features, with a performance peak at 9 seconds pre-error across both modalities potentially marking an early cognitive shift toward error states. Both strategies exhibited characteristic performance decline in the final second before error commission (EEG: 75.0%, eye-tracking: 70.0%), possibly reflecting transition from anticipatory neural activity to overt execution of erroneous behaviour [39].

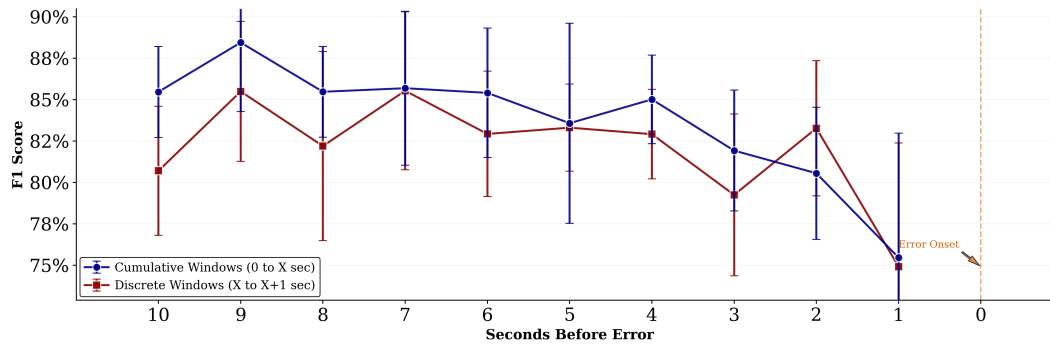


Figure 3: EEG-based classification F1 scores across the 10-second pre-error window, comparing cumulative (0 to X seconds) and discrete (X to X+1 seconds) approaches. Statistical significance was maintained throughout ($p < 0.001$).

3.3 Modality-Specific Characteristics

EEG consistently outperformed eye-tracking in predicting errors across all timepoints, with an average F1 advantage of 10.3% ($p < 0.001$). The largest performance gap appeared between 7–10 seconds before error onset, indicating that EEG signals reveal early cognitive markers well before oculomotor indicators become prominent. Additionally, EEG classification exhibited lower variability (standard deviation $\pm 1.8\%$) compared to eye-tracking ($\pm 2.5\%$), highlighting greater stability for continuous monitoring applications.

Eye-tracking benefited more from cumulative integration than EEG (4.1% vs. 3.2% improvement), suggesting oculomotor measures may require longer temporal windows to reveal predictive patterns, consistent with prior work on fixation dynamics [1]. These results suggest a complementary multimodal monitoring strategy: EEG provides early, stable detection of impending errors, whilst eye-tracking enhances sensitivity as the point of error approaches.

4 Discussion and Conclusion

This work provides the first empirical validation that physiological signals can reliably predict human errors up to 10 seconds before they occur in operational flight environments. EEG-based classifiers achieved peak F1 scores of 88.7% in laboratory conditions and 82.0% during flight operations, substantially extending the 2–5 second anticipation windows reported in previous laboratory studies [7, 39]. Predictive capability remained robust across progressively complex environments, from controlled laboratory settings to demanding 2G manoeuvres, demonstrating that neurophysiological markers withstand motion artefacts, electromagnetic interference, and gravitational stress. Additionally, cumulative temporal integration improved prediction performance by 3.2% on average for EEG, indicating that monitoring systems should incorporate memory of preceding physiological states rather than relying solely on instantaneous measurements.

Modality-specific analyses revealed complementary strengths of EEG and eye-tracking signals. EEG provided earlier and more stable predictions with an average 10.3% F1 advantage ($p < 0.001$), detecting error signatures as early as 9–10 seconds pre-commission, whilst eye-tracking performance peaked closer to error execution (2–4 seconds pre-error), likely reflecting attentional lapses [14]. Both

modalities exhibited synchronous decline in the final second before error execution, suggesting a physiological transition phase where cognitive control mechanisms may fail irreversibly [39]. These findings support an integrated multimodal approach where EEG provides early stable detection and eye-tracking offers confirmation as errors become imminent, enabling graduated intervention protocols.

Our results lay groundwork for cognitive-aware cockpit systems that could dynamically assess pilot cognitive states and provide timely interventions. Beyond aviation, similar approaches could enhance safety in surgery, air traffic control, and autonomous vehicle oversight, and EEG has shown clinical promise in detecting early neurological markers of disease [35]. EEG-based monitoring has also demonstrated broader utility in information retrieval, where neural signals reveal cognitive states such as information need [25, 28] and relevance assessment [33], and have supported novel brain-machine interface paradigms including neural query-based retrieval [26, 29], suggesting that continuous neurophysiological sensing may generalise across any domain requiring real-time cognitive assessment. Several limitations warrant consideration: our sample size ($n=9$ pilots, $n=4$ eye-tracking) reflects operational constraints typical of in-flight research; controlled multitasking scenarios enabled systematic comparison but may not capture full complexity of actual flight operations; temporal clustering of errors was not examined; and validation across diverse aircraft types remains necessary. Additionally, real-world deployment requires addressing system latencies and important ethical considerations regarding data privacy, pilot autonomy, and trust in automation.

This research demonstrates that pre-emptive physiological error detection is feasible in operational aviation environments, moving closer to transforming aviation safety from reactive analysis to proactive intervention. By understanding cognition itself and anticipating vulnerabilities before they manifest, these findings point toward enhanced human-machine collaboration that could prevent errors at their neurophysiological origin.

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